Aff

C1: Grid

Data centers are increasing strain on the grid

Dominici 25 (Francesca, Francesca Dominici is the Clarence James Gamble professor of Biostatistics, Population and Data Science at the Harvard T.H. Chan School of Public Health and director of the Harvard Data Science Initiative at Harvard University, 2-16-25, The Dilemma Between A Strained Power Grid And Sustainable AI, https://www.forbes.com/sites/francescadominici/2025/02/16/the-dilemma-between-a-strained-power-grid-and--sustainable-ai/) DH DOA: 8-13-26

The U.S. power grid is under increasing strain from all angles due to rising electricity demand, aging infrastructure, and the challenges of integrating new energy technologies. This will have real impacts on the quality of life for US citizens. AI data centers alone could account for 44% of U.S. electricity load growth from 2023 to 2028. By 2030, they are expected to account for 11-12% of total U.S. power demand, up from 3-4% today. Residential electricity prices are expected to rise from 15 cents/kWh in 2022 to an average of 16.7 cents/kWh in 2025, reflecting higher demand and infrastructure costs. The transition to EVs is not currently straining the U.S. power grid significantly because most EV charging occurs at night when overall electricity demand is low, which helps balance the load on the grid. However, by 2035, the uptake of EVs is projected to require 3,360% more electricity, which could strain the grid if not adequately managed. High temperatures drive up electricity consumption as people rely heavily on air conditioning. This can causes demand to outstrip supply, especially during prolonged heat waves. Furthermore, heat waves often coincide with other extreme weather conditions, creating multiple threats to grid stability. The Climate Central report stated that of all significant U.S. power outages reported from 2000 to 2023, 80% (1,755) were due to weather. Power loss during extreme heat can be life-threatening, especially for vulnerable populations reliant on cooling systems or medical devices. Concurrent heat waves and power outages can more than double heat-related mortality rates. For example, simulations in three U.S. cities (Atlanta, Detroit, and Phoenix) showed that 3% and 50% of the urban population could require medical attention during such events, depending on location and intensity. Vulnerable groups, such as older adults, individuals with preexisting conditions (e.g., cardiovascular or respiratory diseases), and those relying on electricity-dependent medical devices, face heightened risks. Wildfires can cause extensive damage to the power grid and infrastructure, including burning or destroying wooden utility poles and damaging transmission and distribution lines. They also lead to widespread power outages in affected areas. Faulty grid equipment can spark wildfires, as seen in the 2018 Camp Fire, the costliest wildfire in U.S. history.

Its getting worse

Riley 26 (Jennifer, *regional communications lead for the Americas in the Power Systems division of Rolls-Royce,* 8-5-26, AI's Obstacle: Why the Grid Can't Keep Pace With the Data Center Boom, https://www.powermag.com/ais-obstacle-why-the-grid-cant-keep-pace-with-the-data-center-boom/) DH DOA: 8-13-26

Growth in artificial intelligence (AI) is outpacing the world's ability to generate enough power to sustain it. As AI adoption accelerates worldwide, it's colliding with a power system that wasn't built for its level of demand. Vittorio Pierangeli, senior vice president PowerGen at Rolls-Royce Power Systems (Figure 1), explains why electricity, not chips or algorithms, is the defining factor of the AI era.Measuring AI's Exploding Appetite for Power The scale of the AI shift becomes clear when you look back less than a decade. In 2018, the AI-generated portrait "Edmond de Belamy" made global headlines as a novelty, trained on a modest 5 to 7 gigabytes of data before selling at a Christie's auction for $432,500 . Fast-forward to today: roughly 43 million AI-generated images are produced every single day. The models behind them are trained on billions of data points, as much as 100,000 times the computational load of that early art experiment. And image generation is only a sliver of how AI is now woven into daily life. We've crossed into a new era of power demand, with AI moving well past novelty status to become one of the world's fastest-growing drivers of electricity consumption. The Hitch: Power Availability AI depends on reliable power—and a lot of it. Compare an AI-powered search with a conventional one. Depending on how complex the prompt is, a single ChatGPT query can use 10 to 100 times more energy than a standard Google search. Scale that up to the data center floor, and AI server racks now draw 100 to 120 kW of power apiece, versus roughly 10 kW just five years ago. This increased demand coupled with a changing energy mix and geopolitical uncertainty only compounds the challenges in providing reliable power. "Conventional coal plants are being decommissioned, while renewable generation remains intermittent," said Pierangeli. "In addition, geopolitical instability is increasing pressure on energy security, and the load profiles of AI data centers are growing dramatically more volatile." Projections suggest the entire power generation market will nearly

triple between 2025 and 2030, driven largely by data center demand, with the backup power segment expanding roughly 22% a year and continuous power growing at 24% annually. Supply, however, is not keeping up. "Cumulative U.S. grid power supply to data centers is forecast to fall more than 50 gigawatts short of demand by 2030," said Pierangeli. "And while a data center itself can be built in 18 to 24 months, securing a grid connection takes three to seven years." In the race to support AI demandd, speed to market is imperative and hyperscalers simply cannot afford to wait years for a utility interconnection. That gap is opening the door to solutions, such as Rolls-Royce mtu gas gensets in the near term (Figure 2) and small modular reactors (SMRs) further out. Hyperscaler capital expenditure announcements for data centers jumped more than 50% in early 2026 alone and the utility sector has yet to catch up.

Blackouts will be 100x more likely in the next 4 years

DOE 25 (Department of Energy, 7-7-25, "Department of Energy Releases Report on Evaluating U.S. Grid Reliability and Security") DH DOA: 8-13-26

WASHINGTON— The U.S. Department of Energy (DOE) today released its Report on Evaluating U.S. Grid Reliability and Security. The report fulfills Section 3(b) of President Trump's Executive Order, Strengthening The Reliability And Security Of The United States Electric Grid, by delivering a uniform methodology to identify at-risk regions and guide Federal reliability interventions. The analysis reveals that existing generation retirements and delays in adding new firm capacity, driven by the radical green agenda of past administrations, will lead to a surge in power outages and a growing mismatch between electricity demand and supply, particularly from artificial intelligence (AI)-driven data center growth, threatening America's energy security. "This report affirms what we already know: The United States cannot afford to continue down the unstable and dangerous path of energy subtraction previous leaders pursued, forcing the closure of baseload power sources like coal and natural gas," Secretary Wright said. "In the coming years, America's reindustrialization and the AI race will require a significantly larger supply of around-the-clock, reliable, and uninterrupted power. President Trump's administration is committed to advancing a strategy of energy addition, and supporting all forms of energy that are affordable, reliable, and secure. If we are going to keep the lights on, win the AI race, and keep electricity prices from skyrocketing, the United States must unleash American energy." Highlights of the Report: The status quo is unsustainable. DOE's analysis shows that, if current retirement schedules and incremental additions remain unchanged, most regions will face unacceptable reliability risks within five years and the Nation's electrical power grid will be unable to meet expected demand for AI, data centers, manufacturing and industrialization while keeping the cost of living low for all Americans. Staying on the present course would undermine U.S. economic growth,[and] national security, and leadership in emerging technologies. Grid growth must match the pace of AI innovation. Electricity demand from AI-driven data centers and advanced manufacturing is rising at a record pace. The magnitude and speed of projected load growth cannot be met with existing approaches to load addition and grid management. Radical change is needed to unleash the transformative potential of innovation. With projected load growth, retirements increase the risk of power outages [increases] by 100 times in 2030. Allowing 104 GW of firm generation to retire by 2030—without timely replacement—could lead to significant outages when weather conditions do not accommodate wind and solar generation. Modeling shows annual outage hours could increase from single digits today to more than 800 hours per year. Such a surge would leave millions of households and businesses vulnerable. We must renew a focus on firm generation and continue to reverse radical green ideology in order to address this risk. Planned supply falls short, reliability at risk. The 104 GW of plant retirements are replaced by 209 GW of new generation by 2030; however, only 22 GW comes from firm baseload generation sources. Even assuming no retirements, the model found outage risk in several regions rises more than 30-fold, proving the queue alone cannot close the dependable-capacity deficit. Old tools won't solve new problems. Traditional peak-hour tests to evaluate resource adequacy do not sufficiently account for growing dependence on neighboring grids. At a minimum, modern methods of evaluating resource adequacy need to incorporate frequency, magnitude, and duration of power outages, move beyond exclusively analyzing peak load time periods, and develop integrated models to enable proper analysis of increasing reliance on neighboring grids.

Scenario 1 is Econ

Blackouts devastate the economy — supply disruption reduces consumption, increases prices, and disrupts transportation

Wing et. al 25 (Ian Sue Wing has a PhD from MIT, Peter H. Larsen has a PhD from Stanford, Juan Pablo Carvallo, Alan Sanstad, Dan Wei, Adam Rose, Sunhee Baik, Jeremy Smith, Christopher Ramee & Ridge Peterson. April 8, 2025, "A Method to estimate the economy-wide consequences of widespread, long duration electric power interruptions", https://www.nature.com/articles/s41467-025-58537-4) SV DOA 9-6-26

We partnered with a utility in the U.S. state of Illinois to develop and pilot an approach to estimate the economic impacts of widespread, long duration (WLD) power interruptions. We surveyed their customers about hypothetical blackouts, identifying and classifying mitigating/resilience behaviors and quantifying their costs and benefits. Survey results are scaled up to the broader regional economy, and used to drive a computational general equilibrium (CGE) simulation of the effects of power interruptions and attendant customer responses (e.g., relocation, backup generation). Impacts are severe: 1-, 3-, and 14-day interruptions reduce the utility service area's three-month GDP by $1.8 Bn (1.3%), $3.7 Bn (2.6%) and $15.2 Bn (10.4%), respectively, with losses driven overwhelmingly by disequilibrium responses to shortages as opposed to price signals (71%–88%). Doubling backup power penetration moderates GDP losses by 11%–14%, and is relatively least beneficial during the longest interruption duration. Results highlight previously unquantified economic losses that can potentially be avoided by investments in power system resilience. Virtually all economic activities depend on electric power, making individual and collective vulnerability to service disruptions a major concern. Although most interruptions originate at the distribution system-level and cause minimal disturbance to daily life, widespread and long-duration (WLD) interruptions (affecting entire utility service territories or larger regions, and lasting days, weeks, or longer) can have large social and economic impacts1. Estimating the benefits to ratepayers and society of reducing economic losses associated with power interruptions—especially WLDs—is essential to regulatory and utility planning for power system reliability and resilience. Power interruptions' economic losses emanate from two sources. The first, market equilibrium impacts, arise from curtailment of electricity supply, and hence consumption, below the level demanded in the economy's baseline state. Under conditions of normal market functioning, firms and households respond by substituting other inputs for electricity, which in turn stimulates price and quantity adjustments across the economy. Such responses generally increase electricity-using firms' production costs and the prices of their commodities, reduce factor hiring and remuneration to households, and, as consumers simultaneously face rising prices and declining incomes, reduce households' real consumption and well-being. The second, disequilibrium impacts, arise when power disruptions trigger breakdowns in the normal functioning of markets by disrupting utility services, transportation links, and movements of goods and people. The short-run economic equilibrium consequences are that commodity demands exceed supplies (including inventories), and prices neither reflect true scarcity nor incentivize producers to increase supply to alleviate shortages. Firms and households take actions—uncoordinated by price signals—to preserve their profits and well-being, respectively (e.g., the use of backup generators). Such resilience tactics, although privately beneficial (indeed, they may be vital for survival), incur additional costs that fall directly on the customers themselves and affect prices in unanticipated ways. In such settings, the price and quantity adjustments necessary to restore the economy to its baseline state are likely to diverge from those manifesting in market equilibrium.

Decline causes nuclear war

Stein **Tønnesson 15**, Research Professor, Peace Research Institute Oslo; Leader of East Asia Peace program, Uppsala University, 2015, "Deterrence, interdependence and Sino–US peace," International Area Studies Review, Vol. 18, No. 3, p. 297-311

Several recent works on China and Sino–US relations have made substantial contributions to the current understanding of how and under what circumstances a combination of nuclear deterrence and **economic** interdependence may reduce the risk of war between major powers. At least four conclusions can be drawn from the review above: first, those who say that interdependence may ***both inhibit and drive conflict*** are right. **Interdependence raises the cost of conflict** for all sides but asymmetrical or unbalanced dependencies and ***negative trade expectations*** may generate tensions leading to trade wars

among inter-dependent states that in turn increase the risk of military conflict (Copeland, 2015: 1, 14, 437; Roach, 2014). The risk may increase if one of the interdependent countries is governed by an inward-looking socio-economic coalition (Solingen, 2015); second, the risk of war between China and the US should not just be analysed bilaterally but include their allies and partners. Third party countries could drag China or the US into confrontation; third, in this context it is of some comfort that the three main economic powers in Northeast Asia (China, Japan and South Korea) are all deeply integrated economically through production networks within a global system of trade and finance (Ravenhill, 2014; Yoshimatsu, 2014: 576); and fourth, decisions for war and peace are taken by very few people, who act on the basis of their future expectations. International relations theory must be supplemented by foreign policy analysis in order to assess the value attributed by national decision-makers to economic development and their assessments of risks and opportunities. If **leaders** on either side of the Atlantic begin to seriously fear or anticipate their own nation's decline then they may **blame** this on **external dependence,** appeal to anti-foreign sentiments, contemplate the use of force to gain respect or credibility, adopt protectionist policies, and ultimately **refuse to be deterred by** either **nuclear arms** or prospects of socioeconomic calamities. Such **a dangerous shift could happen abruptly**, i.e. under the instigation of actions by a third party – or against a third party. Yet as long as there is both nuclear deterrence and interdependence, the tensions in East Asia are unlikely to escalate to war. As Chan (2013) says, all states in the region are aware that they cannot count on support from either China or the US if they make provocative moves. The **greatest risk is not** that **a territorial dispute** leads to war under present circumstances but that **changes in the world economy** alter those circumstances in ways that render inter-state peace more precarious. If China and the US fail to rebalance their financial and trading relations (Roach, 2014) then a trade war could result, interrupting transnational production networks, provoking social distress, and exacerbating nationalist emotions. This could have unforeseen consequences in the field of security, with **nuclear deterrence** remaining the only factor to **protect the world from Armageddon**, and **unreliably so**. **Deterrence** could **lose its credibility**: one of the two **great powers** might **gamble** that the other yield **in a** cyber-war or conventional limited **war**, or third party countries might engage in conflict with each other, with a view to obliging Washington or Beijing to intervene.

Scenario 2 is Meltdowns

Grid failure causes existential nuclear accidents.

Leyse '25 [Mark; April 28; Nuclear safety analyst, researcher, and consultant, M.S. in Nuclear Engineering from the University of Wisconsin at Madison; The Guardian, "How a solar storm could lead to a US nuclear disaster worse than Chornobyl," https://www.theguardian.com/commentisfree/2025/apr/28/solar-storm-nuclear-disaster]

**edited for easier reading

In a months-long **blackout**, nuclear plants would lose their supply of offsite **electricity**, which is **necessary** for their **safe operation**. Emergency diesel generators, which provide backup electricity, are designed to power cooling pumps for a number of days – not months. No nuclear plant in the United States has ever lost offsite **electricity** for longer than a **week**. In 2012, the US Nuclear Regulatory Commission stated that an extreme solar storm could collapse power grids and potentially lead to reactor **core damage** at multiple nuclear plants. The nuclear industry's mismanagement of spent fuel must stop. Storage pools at nuclear plants that house spent fuel assemblies, encasing nuclear waste, are also vulnerable to **accidents**. Spent fuel assemblies can **overheat** and catch fire, dispersing **radioactive material** into the **environment**, if a storage pool's coolant water is lost. Spent fuel assemblies are so thermally hot and radioactive that they must be submerged in circulating water and cooled in a storage pool for years. Storage pools at **US** nuclear plants **[npp]** typically contain about six reactor **core loads** of **nuclear fuel** and are almost

as densely packed with fuel assemblies as operating reactors – hazards that vastly increase the odds of a **major accident**. A spent fuel fire in an exposed, densely packed storage pool could potentially release **10 times** as much **caesium-137** as the **Chornobyl** accident is estimated to have released. Such a disaster could contaminate **thousands** of square miles of land and expose millions of people to large doses of ionizing radiation, many of whom might die from early or latent cancer.

Extinction.

Slocum '15 [Christopher Allen; July 12; Vice President of Arizona Oil and Gas Incorporated, J.D. from the University of Missouri at Kansas City; The Hidden Costs of Alternative Energy Series, "A Theory for Human Extinction: Mass Coronal Ejection and Hemispherical Nuclear Meltdown," http://azoilgas.com/wp-content/uploads/2018/03/Theory-for-Human-Extinction-Slocum-20151003.pdf]

With our intelligence we have littered the planet with massive spent nuclear fuel pools, emitting **lethal radiation** in over-crowded conditions, with circulation requirements of **electricity**, **water**-supply, and neutron **absorbent chemicals**. The failure of **any** of these conditions for any calculable or incalculable reason, will release all of a pool's **cesium** into **the atmosphere**, causing 188 square miles to be contaminated, 28,000 cancer deaths and $59 billion in damage. As of 2003, 49,000 tons of SNF was stored at 131 sites with an additional 2,000-2,400 metric tons produced annually. The NRC has issued permits, and the nuclear industry has amassed unfathomable waste on the premise that a deep geological storage facility would be available to remediate the waste. The current chances for a deep geological storage facility look grim. The NAS has required geologic stability for 1,000,000 years. It is impossible to calculate any certainty 1,000,000 years into the future. Humanity could not even predict the mechanical failures at Three Mile Island or Chernobyl, nor could it predict the size of the tsunami that triggered three **criticality** events at Fukushima Daiichi. These irremediable crises span just over 70 years of human history. How can the continued production and maintenance of SNF in pools be anything but a **precedent** to an **unprecedented human cataclysm**? The Department of Energy's outreach website explains nuclear fission for power production, providing a timeline of the industry. The timeline ends, as does most of the world's reactor construction projects in the 1990s, with the removal of the FCMs from Three Mile Island. One would think the timeline would press into the current decade, however the timeline terminates with the question, "How can we minimize the risk? What do we do with the waste?" (The History of Nuclear Energy 12). Nearly fifteen years into the future, these questions are no closer to an answer. The reactors at Fukushima Daiichi are still emitting radioisotopes into the atmosphere, and their condition is unstable. TEPCO has estimated it could take forty years to recover all of the fuel material, and there are doubts as to whether the decontamination effort can withstand that much time (Schneider 72). A detailed analysis of Chernobyl has demonstrated that nuclear fall-out, whether from thermonuclear explosions, spent fuel pool fires, or reactor core criticality events are deleterious to the **food-chain**. **Cesium** and strontium are taken into the **roots** of **plants** and **food** crops, causing direct **human** and **animal** contamination from ingestion, causing cancer, teratogenicity, mutagenesis and death. Vegetation suffers mutagenesis, reproductive loss, and death. Radioactive fields and **forest** floor**s decimate invertebrate** and **rodent variability** and number necessary to supply nature's **food-chain** and **life cycles**. The **flesh and bones** of freshwater and oceanic biota contribute significantly to the total radiation dose in the food-chain. Fresh **water lakes**, **rivers** and **streams** become radioactive. Potable **aquafers** directly underlying SNFs and FCMs are penetrated by downward migration of radioisotopes. **Humans must eat to live. Humans must have water. No human can survive 5 Sv of exposure** to ionizing radiation, many cannot survive exposure to 1 Sv. Realizing the irremediable devastation caused by one thermonuclear warhead, by one Chernobyl, by one Fukushima Daiichi, it remains to be said that the **earth** can handle as many simultaneous loss of coolant failures as nature can create. **Humanity cannot**. It is not good enough to lead by relegating probable human wide extinction phenomena to an appeal to lack of evidence. Policy **cannot** indefinitely **ignore**

responsibility by requiring further study. Nor can leadership idle into **cataclysm** by relying on the largest known natural phenomena of the last 200 years. Permitting construction and continued operation of malefic machinery, based on 200 years of cataclysmic experience is a protocol for calamity. Of coronal mass ejections, Hapgood warns, that we need to prepare for a once-in-1000-year event, not just simulate infrastructure safeties by the measure of what we have seen in the past. The same is true for all natural phenomena. The future of humanity is too precious to operate with such insouciance. The engineering is not good enough. It never will be. Nature is too unpredictable, and nuclear power is too dangerous.

Aff solves. Moratoriums allow policymakers to update infrastructure and regulations

Bommarito 26 [Michael J. Bommarito II1 who went to University of Michigan with at least 7 years of post secondary education, February 15, 2026, "Moratorium Nation: A Survey of Data Center, Renewable Energy, and Battery Storage Moratoria in the United States", https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6242898] AC: DOA – 7/7

Second, moratoria serve as a regulatory catch-up mechanism. In my structured extraction dataset, 64.3% of instruments include a finding of study intent—an express statement that the governing body plans to use the moratorium period to study impacts and develop permanent standards (Section 5.1). Another 39.8% cite a regulatory gap as the basis for the pause. These instruments are not, in the main, permanent prohibitions. They are time-limited responses to a mismatch between the pace of infrastructure development and the pace of local regulatory adaptation. Third, legal authority varies sharply across states. At least thirteen states have enacted express moratorium statutes with defined procedures and duration limits (Section 4.1). The majority rely on implied authority under the police power and home rule provisions, with moratorium validity governed by a reasonableness standard (Section 4.2). Pennsylvania and Virginia prohibit development moratoria categorically, and West Virginia has preempted local moratorium authority for data centers (Section 4.3). States with the most orderly moratorium practice (Wisconsin, Minnesota, Oregon, North Carolina) are those with the most detailed enabling statutes (Wis. Stat. § 66.1002; Minn. Stat. § 462.355; ORS 197.520; N.C.G.S. § 160D-107). Fourth, current drafting practice is uneven. A majority of moratorium instruments (56.1%) provide no formal definition of the regulated use (Section 5.3). Nearly four in five (78.6%) lack exemptions for existing operations, approved projects, or pending applications (Section 5.4). Only 21.4% contain detailed legislative findings. Sector-specific provisions (water assessment, grid impact analysis, fire safety requirements) appear in fewer than one in ten instruments (Section 5.5). These deficiencies expose adopting jurisdictions to constitutional challenge and undermine the effectiveness of the moratorium as a planning tool. Fifth, moratoria cascade. When one jurisdiction enacts a development pause, neighboring jurisdictions frequently follow (Section 6.3). Georgia went from one data center moratorium in March 2025 to at least 14 by February 2026. Michigan saw 26 townships and cities act within a four-month window. Post-Moss Landing BESS moratoria spread from California to New York and beyond within weeks of the January 2025 fire. These cascades can create regional dead zones in which no jurisdiction will accept applications for the target use, amplifying both the economic consequences for developers and the political pressure for state-level legislative response. Sixth, no cross-sector model moratorium template existed before this paper. I surveyed the American Planning Association, the International City/County Management Association, the National League of Cities, and state municipal leagues in every high-moratorium state (Section 7.1). Sector-specific permanent regulatory models exist for BESS, solar, and wind, but none addresses the moratorium instrument itself (American Clean Power Association 2024; New York State Energy Research and Development Authority 2019; Sabin Center for Climate Change Law 2024). The model framework presented in Section 7 and the full template in Appendix C fill this gap. 71 The Moratorium as Democratic Friction Moratoria impose real costs. Project delays, investment uncertainty, and geographic displacement of development are measurable consequences (Section 6.4). One estimate placed blocked or delayed data center investment at $64 billion or more as of early 2025 (Data Center Watch 2025). Solar, wind, and battery storage moratoria slow the deployment of infrastructure needed for the clean energy transition (Powell and O'Neil 2022). These costs fall on developers, ratepayers, and the communities that forgo tax revenue and employment. Yet moratoria also provide something that no other local regulatory tool delivers with comparable speed: a structured pause during which a community can study a novel land use, gather public input, assess infrastructure capacity, and draft tailored standards. The case studies in Section 6.2 show that the moratorium-toregulation pipeline functions when jurisdictions commit resources to the study phase. East Wenatchee, Washington converted a cryptocurrency mining moratorium into permanent performance standards. Coweta County, Georgia replaced its 180-day pause with a permanent data center ordinance within the original window. Social Circle, Georgia moved from moratorium to special use permit requirements in under four months. The appropriate frame is not whether moratoria are categorically desirable or harmful. It is whether they are well-designed and well-executed. A moratorium with detailed findings, a functional definition, the "earlier of" sunset pattern, a phased study plan, and exemptions for vested rights can serve the public interest while minimizing legal exposure and economic disruption. A moratorium adopted reflexively, drafted without definitions, extended repeatedly, and allowed to expire without replacement regulation wastes the time it was meant to buy

C2: Renewables

Data center expansion is reversing the renewable transition — coal plant extensions prove

Yañez-Barnuevo 26 (Miguel Yañez-Barnuevo: graduate degree in urban policy analysis and management from the New School University in New York City, Environmental Energy Study Institute, 1/26/26, "Data Center Buildout Is Hungry for Fossil Fuels", https://www.eesi.org/articles/view/data-center-buildout-is-hungry-for-fossil-fuels) RL - DOA: 9-6-26

Utilities nationwide are receiving an increasing number of requests from data center developers for large amounts of energy to power their facilities. This represents a reversal of a trend since 2009 of relatively flat energy demand and predictable growth. Studies show that the projected electricity demand of data centers in 2028 could reach 130 gigawatts (GW). Data centers could represent up to 12% of total U.S. electricity demand by 2028—up to 580 terawatt-hours per year. Forecasts estimate that overall U.S. electricity demand will reach about 1,250 gigawatts (GW) by 2050, a 78% increase from 2023 levels. About 80 GW of electricity will need to come online every year for the next 20 years to meet that demand. To meet this unprecedented increase in demand, utilities are proposing to extend the life of existing coal power plants by delaying planned retirements, repower unused or mothballed plants, and build new natural gas power plants. Throwing More Coal at the Power Grid Some utilities are moving to extend the life of coal power plants and delaying already-announced facility retirements. The U.S. Department of Energy (DOE) has taken steps to facilitate or even encourage such developments. Invoking the Federal Power Act, DOE has issued several emergency declarations throughout 2025 to stop the retirements of eight coal units across the country, representing a capacity of more than 17 gigawatts. The agency has justified its decisions in the name of grid reliability, to keep electricity prices affordable, and to avoid blackouts. Coal power plants are a reliable, well-tested baseload technology that can operate 365 days per year with limited interruptions. However, that continuous availability comes at a cost: emissions of toxic particulate matter that harms the environment and nearby communities, as well as greenhouse gases that heat the planet. Coal-fired plants also require substantial amounts of water to produce high-temperature steam, which spins turbines and generates electricity. Keeping coal plants running longer than planned would increase emissions and cost ratepayers billions of dollars in higher maintenance costs that are directly reflected in higher bills for consumers. Until recently, a shift in the energy paradigm had been occurring. Over the last decade, as the cost of natural gas and clean, renewable energy fell, coal power plants closed rapidly, resulting in lower carbon emissions. Because coal power is the most polluting source of energy, closing these plants vastly improved air quality and the health of those who live close to them by preventing hundreds of millions of metric tonnes of carbon dioxide and other pollutants from being released into the atmosphere. Between 2000 and 2023, coal's share of the U.S. electric grid declined from 51.7% to 16.2%. The main factor in this dramatic decline was the replacement of coal-fired plants with cheaper natural gas-fired and renewable energy plants. The cost of solar energy declined by 85% between 2010 and 2023, and solar and wind's share of U.S. electricity generation is projected to continue to grow slowly but steadily in the coming years. But now, the skyrocketing energy demands of data centers are giving utilities second thoughts about retiring coal power plants. Nationwide, electric utilities have postponed the already-announced retirements of 15 coal-fired power plants, which, in total, emitted almost 65 million metric tonnes of greenhouse gases in 2023. More un-retirement announcements may come in the next few months, as utilities, grid operators, and the DOE closely reexamine whether coal power assets need to be kept online. For example, in Colorado, retirements of two coal power plants are being reconsidered due to increased energy demand, which could also affect other planned shutdowns in the state. Natural Gas Power Plant Buildout Surging data center development is also prompting utilities to construct new gas-fired power plants. More than 100 GW of new gas-fired facilities have been announced in the last few years, and most are expected to come online before 2030. There was approximately 500 GW of natural gas capacity in the United States in 2022, spread across 2,000 gas-fired plants that emitted 659 million metric tonnes of carbon dioxide. If utilities add 100 GW of new gas facilities, it could potentially result in more than 100 million tonnes of carbon dioxide emissions annually as well as other harmful pollution. This would continue to harm the environment and compromise the health of people living near these new power plants. Extracted methane gas is carried to power plants via pipelines—at a high cost. Constructing new gas-fired power plants entails building expensive new pipelines to supply them. Interstate pipeline construction entails a lengthy permitting process that can take up to seven years due to land acquisition, required environmental assessments, and federal and state approvals, thereby increasing total infrastructure costs. Building these pipelines to deliver natural gas to plants powering data centers can have significant environmental impacts, including contamination of drinking water, soil erosion,

air pollution, and loss of wildlife habitat. Leaks of methane—a superpollutant and potent greenhouse gas—from pipelines and compressor stations pose a threat to the climate and nearby communities.

They’re canceling renewables contracts in favor of gas — speed and cost is all they care about

Varanasi 26 (Lakshmi, writes about AI and the consulting industry, exploring how artificial intelligence is transforming consulting firms like McKinsey & Company, Boston Consulting Group, and Bain. Previously, she reported for Slate, Politico, and Reveal, 5-11-26, “The AI industry is turning to an old energy favorite to bring its data centers online fast” https://www.businessinsider.com/ai-data-center-energy-natural-gas-renewables-environment-2026-4) MG - DOA: 9-6-26

As AI companies race to secure power for their data centers, natural gas is having a moment.

Data centers are growing bigger and more power-intensive. They are being built on tighter timelines than ever before, and the current power infrastructure in the United States is struggling to keep up. "Speed matters," Vivian Lee, a managing director and partner at Boston Consulting Group, told Business Insider. "Bringing a site online even a year earlier can have a meaningful economic impact." Lee said it typically takes two to three years to build a data center, assuming the local community is on board. If they aren't, it could take longer. At the same time, grid upgrades can take four to eight years, so AI companies are looking for faster ways to secure power, Lee said. One of the quickest ways to do that these days is to use existing natural gas infrastructure. Gas plants can often be built or expanded faster than nuclear projects, plug into an extensive pipeline network, and provide greater energy security than renewable resources. Meta is adding seven natural gas plants to its Hyperion data center in Louisiana. Chevron and Engine No. 1, which announced a partnership last year to build natural gas-powered plants next to US data centers, recently brought Microsoft into the fold and are now working to power a data center campus in West Texas. Google has also confirmed that it is working with Crusoe Energy to build natural gas plants to partially power its "Goodnight" data center campus in the Texas Panhandle. "The most important metric now is speed to power — and a lot of it. That's why gas is back in focus," Jamie Webster, a senior director and partner at BCG, said. Renewables take a back seat Natural gas produces less carbon dioxide per unit of energy than coal or oil when burned, but it is still a fossil fuel and a driver of the climate crisis. That makes Silicon Valley's recent embrace of natural gas notable. American tech leaders have long positioned themselves as leaders in the shift to renewables. Tech giants, including Google, Amazon, Microsoft, and Meta, signed massive wind and solar power deals as recently as last year to offset the growing electricity demand of their data centers. Those deals, however, may have been more about money than principle. Webster said the focus on renewables has been driven not only by sustainability goals, but also by their ability to lower energy costs over time, since they're the only energy source whose costs tend to decline. "Over the past decade, technologies like solar, wind, and batteries saw cost reductions of up to 90%, which fundamentally changed the equation," he said. The bottom line is also what's now driving the shift back to natural gas, he said, especially as AI companies raise massive amounts of capital to build the technology's infrastructure while still showing little in comparative revenue. In conversations with developers and energy providers, Lee said renewables are seen as "essential, but not sufficient on their own." Carbon capture technology could help blunt the environmental impact of more natural gas use. The process traps carbon dioxide from power plants or industrial facilities before it reaches the atmosphere, then stores it underground or reuses it. That could allow companies to keep using natural gas while reducing the emissions that make it a major contributor to the climate crisis. Webster, however, said that carbon capture technology — sometimes known as CCUS, or "Carbon Capture, Utilization, and Storage" — is still in the early stages of scaling. The world has now entered a "structural supercycle," driven in part by data centers, electrification, and cooling demand, Webster said."That growth is putting pressure on supply, and in many cases, gas is one of the fastest ways to meet it."

The aff forces data centers to transition to renewable energy

Kyc 26 Kyc, Brandon. "New York State Legislature passes bill to put a one-year moratorium on data centers." Washington-Baltimore News Guild, 05 June 2026, https://www.wbng.com/2026/06/05/new-york-state-legislature-passes-bill-put-one-year-moratorium-data-centers/?outputType=amp.

New York lawmakers have approved a bill that would place new environmental, utility, labor and community benefit requirements on data centers as the industry faces growing scrutiny over its energy and water use. The Responsible Data Center Development Act has passed the state Senate and Assembly and now awaits action by Gov. Kathy Hochul. If signed, it would take effect immediately, and New York would be the first state in the nation to enact such a bill. The bill would create a **one-year pause on certain new permits for large data center**s, defined as facilities with a peak demand of 20 megawatts or more. The **moratorium** would not apply to previously issued approvals or projects that began construction before the law takes effect. The measure would also **require large data center** operators **to** hold at least one in-person public hearing in a host community at least three months before receiving state approval. Residents would have to receive at least 30 days' notice, including information on the project's location, expected energy use, water and wastewater impacts, and any state or local incentives sought or awarded. The bill directs the state Department of Environmental Conservation to prepare a statewide environmental impact report on data center development. The report would examine electricity and water use, land use, pollution, electronic waste, public subsidies, and impacts on disadvantaged communities and Indigenous nations within 10 miles of data centers. A final report would be due within 18 months of the bill becoming law. **The legislation also seeks to prevent costs tied to large data centers from being shifted onto other utility customers.** Electric, gas, water and municipal systems would have to create separate service classifications for large data centers, assigning costs such as infrastructure upgrades and commodity price increases to those facilities. Data centers with a peak load of at least 5 megawatts would have to **meet renewable energy standards beginning in 2030**. They would **need to show that** at least **one-third of their electricity comes from renewable energy from 2030** through 2034, two-thirds from 2035 through 2039, **and 90% starting in 2040. Large data centers would also be required to fund host community benefit programs, which could include residential heat pumps, solar panels, battery storage, broadband upgrades, drinking water infrastructure, noise mitigation, and pollution reduction.**

The bill includes labor standards for data center construction, including prevailing wage, apprenticeship or workforce training requirements, and domestic iron and steel provisions for data centers with a peak demand of 5 megawatts or more.

That's key to public decarbonization

Flanagan 23 (Ciaran, Vice President and Head of Global Data Center Solutions at Siemen, 3-13-23, "How Data Centers Are Driving The Renewable Energy Transition" https://www.forbes.com/sites/siemens-smart-infrastructure/2023/03/13/how-data-centers-are-driving-the-renewable-energy-transition/) MG - DOA: 8-12-26

With the backdrop of a global energy crisis, it's perhaps a good time to discuss the real energy cost of data centers, why they've become one of the world's fastest-advancing industries, and how an argument could be made that they are, in fact, a major driving force in the energy transition to renewables.Firstly, let's consider their growth. Between 2015 and 2021, internet users have increased by 60%, and internet traffic by 440%, according to IEA data. But despite a rise in workload for data centers of 260% in the same time period, global energy use attributed to data centers has remained relatively flat, representing just a 10% increase. Decoupling service demand from energy This decoupling of energy usage from service demand is hugely significant. It's being driven by rapid innovation of digital technology in data centers, allowing the industry to offset huge increases in demand with improved infrastructure efficiency. In short, we are building more and more digital capacity, modernizing legacy infrastructure and meeting the growth, but in real terms we're not using much more energy than we were five years ago. The real challenge here is how we limit the environmental impact of global internet use. We should be honest; there is an impact, there always will be and we need to continue to use technology and commercial innovation to mitigate it. Looking at the lifecycle of a datacenter, we

see that more than 85% of the carbon impact today comes from operations. There is also certainly an impact during construction and it's also significant - certainly worthy of mitigation - but net-zero construction will not address the larger part of the challenge: the carbon impact of running the facility for 20 years or more. Driving the energy transition Enter the concept of energy mix. A data center's carbon performance is broadly a function of the energy mix in the location in which it's operating. There are some exceptions, where operators take the responsibility to generate power on-site using renewables or gas, but largely speaking local grids power data centers. The data center industry is a major buyer of PPA agreements for renewable energy, and this has a significant impact(s) on the energy mix. In 2021, Amazon and Microsoft were the two largest corporate buyers of renewable energy through PPA; to a degree, the data center industry is helping to drive decarbonization by underwriting a significant proportion of grid-scale, carbon-free energy for industry. This can catalyze a whole ecosystem at a national level, and also demonstrates to the broader industrial base that critical loads can reliably move to renewables. Growing adoption of renewable energy - and higher levels of infrastructure redundancy at the IT level - are also leading to new design best practices in data centers. Battery Energy Storage Systems (BESS) are replacing diesel gensets as short-term backup power supply, for example. With energy markets increasingly interconnected, data centers adopting BESS can generate unprecedented revenues by running in island mode in case of outages (without emitting carbon dioxide from diesel generators) or stabilizing the frequency of the grid.

Fossil fuels in power generation are the #1 driver of climate change

Johns Hopkins 25 (Johns Hopkins University School of Advanced International Studies, Master of Arts in Sustainable Energy Program, 7-21-25, "The Role of the Sustainable Energy Transition in Tackling Climate Change" https://energy.sais.jhu.edu/articles/the-future-of-sustainable-energy/) MG - DOA: 8-12-26

The biggest contributor to global climate change is fossil fuels: burning coal, oil, and gas accounts for over 75% of global greenhouse gas emissions and almost 90% of carbon dioxide (CO2) emissions. The biggest use of fossil fuels is to create energy that is used for power, manufacturing, and transportation, which mean that energy production using fossil fuels is the #1 reason for greenhouse gas emissions. Given that our energy production processes are one of the primary reasons for climate change, these are the areas that are most critical for us to change. Successful sustainable energy transitions, especially on the national or even global level, are a key method for humanity to reduce its impact on the environment. Reducing our reliance on fossil fuels, and shifting to renewable sources that harness sustainable infrastructure, will greatly reduce the amount of greenhouse gas and CO2 emissions. How Does Our Reliance on Fossil Fuels Affect The Planet? Human activity, primarily power generation, deforestation, food production, transportation, and overconsumption are driving global warming and other indicators of climate change. As discussed, the reliance on fossil fuels for energy creation is one of the key drivers of climate change. Some of the ways in which our use of fossil fuels impact the planet include: Rising greenhouse gas (GHG) emissions: Fossil fuels represented 81% of the total global energy supply in 2022, with the United States and China comprising 45% of global fuel combustion emissions. Greenhouse gases trap heat in Earth's atmosphere, contributing to higher global temperatures. Rising CO2 emissions: When we burn fossil fuels, carbon dioxide (and other greenhouse gases) are released into the air. While carbon dioxide doesn't trap heat as much as other greenhouse gasses, there is now so much of it in the atmosphere that it accounts for 82% of GHG emissions10. High CO2 concentrations can also lead to a more acidic ocean, harming marine life. By relying on fossil fuels rather than sustainable sources, we're actively contributing to air and water pollution, habitat destruction, and climate change as a whole. By transitioning to renewable sources and sustainable infrastructure, we can make a critical step toward the fight against climate change.

Warming causes extinction—impact magnifier and laundry list

Kemp '22 [Luke, Chi Xu, Joanna Depledge, Kristie Ebi, Goodwin Gibbins, Timothy Kohler, Johan Rockström, Marten Scheffer, Hans Schellnhuber, Will Steffen, Timothy Lenton, and Kerry Emanuel; August 1st; PhD in Political Science and International Relations, Bachelors in Political Science and

Environmental Policy, Fellow at Notre Dame University; Faculty at Nanjing University, PhD; PhD from University College London, Lecturer at Cambridge University, Editor of the journal Climate Policy; Professor in Global Health and Environment and Health Sciences, PhD and MPH at the University of Michigan, MS at MIT, BS at MSU; PhD at Imperial College London, MS at Cambridge; Regents Professor Emeritus at Washington State University, MA and PhD at the University of Florida; Professor in environmental science, Ph.D. in ecology from Stockholm University; Professor, PhD at Utrecht University; Ph.D. in Theoretical Physics, Director, Potsdam Institute for Climate Impact Research; Emeritus Professor Fenner School of Environment & Society at Australia National University, MSc and PhD from the University of Florida; PhD, Professor at the University of Exter; co-director of the Lorenz Center at MIT, Professor, Department of Earth, Atmospheric and Planetary Sciences, Massachusetts Institute of Technology, "Climate Endgame: Exploring catastrophic climate change scenarios,", Centre for the Study of Existential Risk, University of Cambridge, https://www.pnas.org/doi/10.1073/pnas.2108146119]

Worst-Case Climate Change Despite 30 y of efforts and some progress under the United Nations Framework Convention on Climate Change (UNFCCC) anthropogenic greenhouse gas (GHG) emissions **continue** to **increase**. Even without considering worst-case climate responses, the current trajectory puts the world on track for a temperature rise between 2.1 °C and 3.9 °C by 2100 (11). If all 2030 nationally determined contributions are fully implemented, warming of 2.4 °C (1.9 °C to 3.0 °C) is expected by 2100. Meeting all long-term pledges and targets could reduce this to 2.1 °C (1.7 °C to 2.6 °C) (12). Even these optimistic assumptions lead to **dangerous** Earth system **trajectories** Temperatures of more than 2 °C above preindustrial values have **not been sustained** on Earth's surface since before the Pleistocene Epoch (or more than 2.6 million years ago) (13). Even if anthropogenic GHG emissions start to decline soon, this does not rule out high future GHG concentrations or extreme climate change, particularly beyond 2100. There are **feedbacks** in the carbon cycle and potential **tipping points** that could generate high GHG concentrations (14) that are often missing from models. Examples include Arctic **permafrost thawing** that releases methane and CO2 (15), carbon loss due to **intense droughts** and fires in the Amazon (16), and the apparent slowing of dampening feedbacks such as natural **carbon sink** capacity (17, 18). These are likely to not be proportional to warming, as is sometimes assumed. Instead, abrupt and/or irreversible changes may be **triggered** at a **temperature threshold**. Such changes are evident in Earth's geological record, and their impacts cascaded across the coupled climate–ecological–social system (19). Particularly worrying is a "tipping cascade" in which multiple tipping elements interact in such a way that **tipping one** threshold **increases** the **likelihood** of tipping **another** (20). Temperature rise is crucially dependent on the overall dynamics of the Earth system, not just the anthropogenic emissions trajectory. The potential for tipping points and higher concentrations despite lower anthropogenic emissions is evident in existing models. Variability among the latest Coupled Model Intercomparison Project Phase 6 (CMIP6) climate models results in overlap in different scenarios. For example, the top (75th) quartile outcome of the "middle-of-the-road" scenario (Shared Socioeconomic Pathway 3-7.0, or SSP3-7.0) is substantially hotter than the bottom (25th) quartile of the highest emissions (SSP5-8.5) scenario. Regional temperature differences between models can exceed 5 °C to 6 °C, particularly in polar areas where various tipping points can occur (SI Appendix). There are even more uncertain feedbacks, which, in a very worst case, might amplify to an **irreversible transition** into a **"Hothouse Earth"** state (21) (although there may be negative feedbacks that help buffer the Earth system). In particular, poorly understood cloud feedbacks might trigger sudden and irreversible global warming (22). Such effects remain underexplored and largely speculative "unknown unknowns" that are still being discovered. For instance, recent simulations suggest that stratocumulus cloud decks might abruptly be lost at CO2 concentrations that could be approached by the end of the century, causing an additional ∼8 °C global warming (23). Large uncertainties about dangerous surprises are reasons to prioritize rather than neglect them. Recent findings on equilibrium climate sensitivity (ECS) (14, 24) underline that the magnitude of climate change is uncertain even if we knew future GHG concentrations. According to the IPCC, our best estimate for ECS is a 3 °C temperature rise per doubling of CO2, with a "likely" range of (66 to 100% likelihood) of 2.5 °C to 4 °C. While an ECS below 1.5 °C was essentially ruled out, there remains an 18% probability that ECS could be greater than 4.5 °C (14). The distribution of ECS is "heavy tailed," with a higher probability of very high values of ECS than of very low values. There is significant uncertainty over future anthropogenic GHG emissions as well. Representative Concentration Pathway 8.5 (RCP8.5, now SSP5-8.5), the highest emissions pathway used in IPCC scenarios, most closely matches cumulative emissions to date (25). This may not be the case going forward, because of falling prices of renewable energy and policy responses (26). Yet, there remain reasons for caution. For instance, there is significant uncertainty over key variables such as energy demand and economic growth. Plausibly higher economic growth rates could make RCP8.5 35% more likely (27). Why Explore Climate Catastrophe? Why do we need to know about the plausible worst cases? First, risk management and robust decision-making under uncertainty requires knowledge of extremes. For example, the minimax criterion ranks policies by their worst outcomes (28). Such an approach is particularly appropriate for areas characterized by high uncertainties and tail risks. Emissions trajectories, future concentrations, future warming, and future impacts are all characterized by uncertainty. That is, we can't objectively prescribe probabilities to different outcomes (29). Climate damages lie within the realm of "deep uncertainty": We don't know the probabilities attached to different outcomes, the exact chain of cause and effect that will lead to outcomes, or even the range, timing, or desirability of outcomes (, 30). Uncertainty, deep or not, should motivate precaution and vigilance, not complacency. Catastrophic impacts, even if unlikely, have major implications for economic analysis, modeling, and society's responses (31, 32). For example, extreme warming and the consequent damages can significantly increase the projected social cost of carbon (31). Understanding the vulnerability and responses of human societies can inform policy making and decision-making to prevent systemic crises. Indicators of key variables can provide early warning signals (33). Knowing the worst cases can compel action, as the idea of "nuclear winter" in 1983 galvanized public concern and nuclear disarmament efforts. Exploring severe risks and higher-temperature scenarios could cement a recommitment to the 1.5 °C to 2 °C guardrail as the "least unattractive" option (34). Understanding catastrophic climate scenarios can also inform policy interventions, including last-resort emergency measures like solar radiation management (SRM), the injection of aerosols into the stratosphere to reflect sunlight (35). Whether to resort to such measures depends on the risk profiles of both climate change and SRM scenarios. One recent analysis of the potential catastrophic risk of stratospheric aerosol injection (SAI) found that the direct and systemic impacts are under-studied (36). The largest danger appears to come from "termination shock": abrupt and rapid warming if the SAI system is disrupted. Hence, SAI shifts the risk distribution: The median outcome may be better than the climate change it is offsetting, but the tail risk could be worse than warming (36). There are other interventions that a better understanding of catastrophic climate change could facilitate. For example, at the international level, there is the potential for a "tail risk treaty": an agreement or protocol that activates stronger commitments and mechanisms when early-warning indicators of potential abrupt change are triggered.

The Potential for Climate Catastrophe

There are four key reasons to be concerned over the potential of a **global climate catastrophe**. First, there are warnings from history. Climate change (either regional or global) has played a role in the **collapse** or transformation of numerous **previous societies** (37) and in each of the **five mass extinction events** in Phanerozoic Earth history (38). The current **carbon** pulse is occurring at an **unprecedented** geological **speed** and, by the end of the century, may **surpass thresholds** that **triggered previous** mass **extinctions** (39, 40). The worst-case scenarios in the IPCC report project temperatures by the 22nd century that last prevailed in the Early Eocene, reversing 50 million years of cooler climates in the space of two centuries (41).

This is particularly alarming, as human societies are locally adapted to a specific climatic niche. The rise of large-scale, urbanized agrarian societies began with the shift to the stable climate of the Holocene ∼12,000 y ago (42). Since then, human population density peaked within a narrow climatic envelope with a mean annual average temperature of ∼13 °C. Even today, the most economically productive centers of human activity are concentrated in those areas (43). The cumulative impacts of warming may **overwhelm societal adaptive capacity.**

Second, climate change could directly trigger other catastrophic risks, such as **international conflict**, or exacerbate **infectious disease spread**, and spillover risk. These could be **potent extreme threat multipliers.**

Third, climate change could exacerbate vulnerabilities and cause multiple, indirect stresses (such as economic damage, loss of land, and **water** and **food insecurity**) that coalesce into **system-wide synchronous failures**. This is the path of systemic risk. Global crises tend to occur through such reinforcing "synchronous failures" that **spread across countries** and systems, as with the 2007–2008 global financial crisis (44). It is plausible that a sudden shift in climate could trigger systems failures that **unravel societies across the globe.**

The potential of systemic climate risk is marked: The most vulnerable states and communities will continue to be the hardest hit in a warming world, exacerbating inequities. Fig. 1 shows how projected population density intersects with extreme >29 °C mean annual temperature (MAT) (such temperatures are currently restricted to only 0.8% of Earth's land surface area). Using the medium-high scenario of emissions and population growth (SSP3-7.0 emissions, and SSP3 population growth), by 2070, around 2 billion people are expected to live in these extremely hot areas. Currently, only 30 million people live in hot places, primarily in the Sahara Desert and Gulf Coast (43).

Fig. 1.

Overlap between future population distribution and extreme heat. CMIP6 model data [from nine GCM models available from the WorldClim database (45)] were used to calculate MAT under SSP3-7.0 during around 2070 (2060–2080) alongside Shared SSP3 demographic projections to ~2070 (46). The shaded areas depict regions where MAT exceeds 29 °C, while the colored topography details the spread of population density.

Extreme temperatures combined with high humidity can negatively affect outdoor worker productivity and yields of major cereal crops. These deadly heat conditions could significantly affect populated areas in South and southwest Asia(47).

Fig. 2 takes a political lens on extreme heat, overlapping SSP3-7.0 or SSP5-8.5 projections of >29 °C MAT circa 2070, with the Fragile States Index (a measurement of the instability of states). There is a striking overlap between currently vulnerable states and future areas of extreme warming. If current political fragility does not improve significantly in the coming decades, then a belt of instability with potentially serious ramifications could occur.

Fig. 2.

Fragile heat: the **overlap between state fragility**, **extreme heat**, and **nuclear and biological catastrophic hazards**. GCM model data [from the WorldClim database (45)] was used to calculate mean annual warming rates under SSP3-7.0 and SSP5-8.5. This results in a temperature rise of 2.8 °C in ~2070 (48) for SSP3-7.0, and 3.2 °C for SSP5-8.5. The shaded areas depict regions where MAT exceeds 29 °C. These projections are overlapped with the 2021 Fragile State Index (FSI) (49). This is a necessarily rough proxy because FSI only estimates current fragility levels. While such measurements of fragility and stability are contested and have limitations, the FSI provides one of the more robust indices. This Figure also identifies the capitals of states with nuclear weapons, and the location of maximum containment Biosafety Level 4 (BS4) laboratories which handle the most dangerous pathogens in the world. These are provided as one rough **proxy for nuclear and biological catastrophc hazards**.

Finally, climate change could **irrevocably undermine humanity's ability to recover from another cataclysm**, **such as nuclear war**. That is, it could create significant latent risks (Table 1): Impacts that may be manageable during times of stability become dire when responding to and recovering from catastrophe. These different causes for catastrophic concern are interrelated and must be examined together.

AND, it triggers knock-on effects—Extinction.

Wright '22 [Gwyn; August 1; Science reporter, citing Climate Endgame, a peer-reviewed entry published in the proceedings of the National Academy of Sciences by Dr. Luke Kemp, Dr. Chi Xu, Dr. Kristie Ebi, Dr. Goodwin Gibbins, Dr. Timothy Kohler, Dr. Johan Rockström, Dr. Martin Scheffer, Dr. Hans Joachim Schellenhuber, Dr. Will Steffen, and Dr. Timothy Lenten; Study Finds, "'Dangerously

underexplored': Experts warn climate change could lead to extinction of humanity," https://studyfinds.org/climate-change-extinction-humanity/]

The world is on track for **5.2°F** of warming by the year **2100** if governments' existing polices — as opposed to pledges they have made — continue without change, according to Climate Action Tracker. Scientists say 2.7°F is a safe level of heating.

The 'four horsemen' of climate change

The team is calling for new research to focus on the "four horsemen" of the climate endgame: **famine** and malnutrition, **extreme weather**, **conflict**, and vector-borne **diseases**.

Rising temperatures increase the risk of crop failures in the world's most fertile agricultural areas. Meanwhile, hotter weather could cause outbreaks of deadly new diseases as habitats for both people and animals shift and shrink.

The authors say catastrophic warming will also make other **existing** threats **worse**, including rising inequality, **misinfo**rmation, **democratic breakdown**, and even new forms of destructive **AI weaponry**.

They add that **tech**nologically advanced **superpowers** may end up fighting each other in "**warm wars**," where they fight over dwindling carbon space and even fund expensive experiments to deflect sunlight and reduce global temperatures.

Researchers need to better understand tipping points that could spark disaster, such as melting permafrost that releases methane, the loss of forests that store carbon, and even cloud cover, the team argues.

'Climate change has played a role in every mass extinction event'

They are calling on the Intergovernmental Panel on Climate Change (IPCC) to compile a report on catastrophic climate change in a bid to galvanize research and better inform the public.

"There are **plenty of reasons** to believe climate change could **be**come catastrophic, even at **modest levels** of warming," says lead author Dr. Luke Kemp from Cambridge's Centre for the Study of Existential Risk in a media release.

"Climate change has played a role in every mass extinction event. It has helped fell empires and shaped history. Even the modern world seems adapted to a particular climate niche."

"Paths to disaster are not limited to the direct impacts of high temperatures, such as extreme weather events. **Knock-on** effects such as **financial crises**, conflict, and new **disease outbreaks** could trigger other calamities, and impede **recovery** from potential disasters such as **nuclear war**," Dr. Kemp continues.